Linear Models Lecture 3: Estimation of the Linear Projection Model

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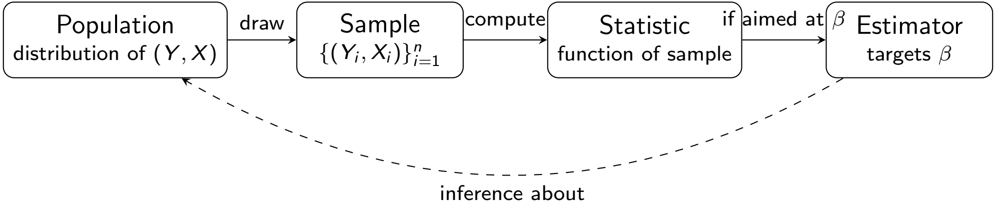
March 30, 2026

Deriving the OLS estimator

- Recall, geometry can work in populations or samples.
- In the following slides we will derive
  - the OLS slope estimator $\hat{\beta}$,
  - the OLS estimator for error variance,
  - Decomposition of OLS (Analysis of Variance), with R^2
- Next time, the FWL Theorem.

Review: From Population to Inference

- The CEF and linear projection are facts about **populations**—we derived $\beta = [\mathbb{E}[XX']]^{-1}\mathbb{E}[XY]$ last time.
- But we don't observe populations. We observe **samples**.



- A statistic is any function of the sample (e.g. $\bar{Y}$, s^2, a histogram).
- An estimator is a statistic designed to estimate a population parameter.

Samples

- We estimate the projection coefficient β with measurements of (Y, X), a *sample*.
- $\{(Y_i, X_i) : \ i \in 1, \ldots, \ n\}$, is the sample, governed by the distribution of (Y, X).
- From the next slide forward, $\boldsymbol{X}$ is going to be a matrix, and we will use lower case for vectors e.g. $\boldsymbol{e}$.

Identically Distributed.

- Main approach is to assume homogeneity: $\{(Y_i, \boldsymbol{x}_i) : \; i \in 1, \ldots, \; n\}$ are identically distributed.

$$
\begin{aligned}
\begin{bmatrix} Y_1 \\ \vdots \\ Y_n \end{bmatrix} &= \beta_1 \begin{bmatrix} 1 \\ \vdots \\ 1 \end{bmatrix} + \beta_2 \begin{bmatrix} X_{21} \\ \vdots \\ X_{2n} \end{bmatrix} + \beta_3 \begin{bmatrix} X_{31} \\ \vdots \\ X_{3n} \end{bmatrix} + \begin{bmatrix} e_1 \\ \vdots \\ e_n \end{bmatrix} \\
&= \begin{bmatrix} 1 & X_{21} & X_{31} \\ \vdots & \vdots & \vdots \\ 1 & X_{2n} & X_{3n} \end{bmatrix} \begin{bmatrix} \beta_1 \\ \beta_2 \\ \beta_3 \end{bmatrix} + \begin{bmatrix} e_1 \\ \vdots \\ e_n \end{bmatrix} \\
\boldsymbol{y} &= \boldsymbol{X}\beta + \boldsymbol{e}
\end{aligned}
$$

- Note, $\mathbb{E}[Y_1|\boldsymbol{x}_1] \neq \mathbb{E}[Y_2|\boldsymbol{x}_2]$ in general (different covariates imply different conditional means).

Least Squares Estimator

- The linear projection coefficient β is the minimizer of the expected squared error

$$S(\beta) = \mathbb{E}[(Y - \boldsymbol{x}'\beta)^2]$$

- The *moment estimator* of $S(\beta)$ is the sample average:

$$\hat{S}(\beta) = \frac{1}{n}\sum_{i=1}^{n}(Y_i - \boldsymbol{x}_i'\beta)^2$$

- $\sum_{i=1}^{n}(Y_i - \boldsymbol{x}_i'\beta)^2$ is called SSE(β), the **sum of squared errors**.
- The estimator $\hat{\beta}$ is the minimizer of $\hat{S}(\beta)$, as well as the minimizer of the SSE.
- $\hat{\beta}$ is called the least squares estimator, sometimes written $\hat{\beta}_n$.

The Plan

- We have a population quantity β and a sample objective $\hat{S}(\beta)$. Now we need to:
  1. **Find** $\hat{\beta}$: minimize SSE using matrix calculus.
  2. **Understand** $\hat{\beta}$ geometrically: projection matrices $\boldsymbol{P}$ and $\boldsymbol{M}$.
  3. **Quantify uncertainty**: estimate σ^2 from residuals, then get standard errors for $\hat{\beta}$.
- Note, residuals $\neq$ disturbances. The former is estimated.

Finite Sample Assumptions of OLS estimator

- Assumption 1: The random variables $\{(Y_1,\ X_1),\ (Y_2,\ X_2),\ldots,(Y_n,\ X_n)\}$ are independent and identically distributed.
- Assumption 2: $Y = \boldsymbol{X}'\beta + e$, where $\mathbb{E}(e|\boldsymbol{X}) = 0$.
- Assumption 3: $\mathbb{E}(\boldsymbol{X}\boldsymbol{X}') > 0$ is invertible (with probability 1).
- Assumption 4*: $\mathbb{E}[e^2|\boldsymbol{X}] = \sigma^2(\boldsymbol{X}) = \sigma^2$.

Derivation of Least Squares Estimator

- The functional form of the least squares estimator can be derived via multivariate calculus.
- We look for a minimum of SSE(β).
- Take first derivatives to calculate first-order conditions and locate the critical values.
- Then we check we have the global minimum, by using the fact that positive definite matrices represent convex functions.

Calculus Derivation of Least Squares Estimator

$$
\begin{aligned}
SSE(\beta) &= (\boldsymbol{y} - \boldsymbol{X}\beta)'(\boldsymbol{y} - \boldsymbol{X}\beta) && \text{(Definition of } \boldsymbol{e}\text{)} \\
&= (\boldsymbol{y}' - \boldsymbol{\beta}'\boldsymbol{X}')(\boldsymbol{y} - \boldsymbol{X}\beta) && \text{(Transpose rules)} \\
&= \boldsymbol{y}'\boldsymbol{y} - \boldsymbol{\beta}'\boldsymbol{X}'\boldsymbol{y} - \boldsymbol{y}'\boldsymbol{X}\beta + \boldsymbol{\beta}'\boldsymbol{X}'\boldsymbol{X}\beta && \text{(Distributive property)} \\
\min_{\boldsymbol{\beta}} SSE(\beta) &= \min_{\boldsymbol{\beta}} \underbrace{\boldsymbol{y}'\boldsymbol{y}}_{\text{constant}} - \underbrace{2\boldsymbol{y}'\boldsymbol{X}\beta}_{\text{linear in } \boldsymbol{\beta}} + \underbrace{\boldsymbol{\beta}'\boldsymbol{X}'\boldsymbol{X}\beta}_{\text{quadratic form}}
\end{aligned}
$$

Calculus Derivation of Least Squares Estimator

$$
\begin{aligned}
\frac{\partial \boldsymbol{y}'\boldsymbol{y} - \boldsymbol{\beta}'\boldsymbol{X}'\boldsymbol{y} - \boldsymbol{y}'\boldsymbol{X}\boldsymbol{\beta} + \boldsymbol{\beta}'\boldsymbol{X}'\boldsymbol{X}\boldsymbol{\beta}}{\partial \boldsymbol{\beta}} &= \mathbf{0} - \boldsymbol{X}'\boldsymbol{y} - \boldsymbol{X}'\boldsymbol{y} + (\boldsymbol{X}'\boldsymbol{X} + \boldsymbol{X}'\boldsymbol{X})\hat{\boldsymbol{\beta}} \\
&= -2\boldsymbol{X}'\boldsymbol{y} + 2\boldsymbol{X}'\boldsymbol{X}\hat{\boldsymbol{\beta}} && \text{(Symmetric)} \\
-2\boldsymbol{X}'\boldsymbol{y} + 2\boldsymbol{X}'\boldsymbol{X}\hat{\boldsymbol{\beta}} &= \mathbf{0} && \text{(find critical point)} \\
\boldsymbol{X}'\boldsymbol{X}\hat{\boldsymbol{\beta}} &= \boldsymbol{X}'\boldsymbol{y}. && \text{(normal equations)} \\
\hat{\boldsymbol{\beta}} &= (\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}'\boldsymbol{y} && \text{(If } \boldsymbol{X}'\boldsymbol{X} \text{ can be inverted (A3).)}
\end{aligned}
$$

Second order condition

Take second derivative, and check that it is positive

$$\frac{\partial - 2\boldsymbol{X}'\boldsymbol{y} + 2\boldsymbol{X}'\boldsymbol{X}\hat{\beta}}{\partial\hat{\boldsymbol{\beta}}'} = 2(\boldsymbol{X}'\boldsymbol{X})'$$

Which is positive so long as $\boldsymbol{X}'\boldsymbol{X}$ is positive definite (Assumption A3).
$\hat{\beta}$ minimizes the sum of squared errors, assuming $\boldsymbol{X}$ is of full rank.

OLS in R: The Formula in Action

```
library(carData)
X <- cbind(1, Prestige$education, Prestige$income)
y <- Prestige$prestige

# The matrix formula from the slides:
solve(t(X) %*% X) %*% t(X) %*% y

# What lm() computes:
coef(lm(prestige ~ education + income, data=Prestige))
```

The two are identical: `lm()` implements $\hat{\beta} = (\boldsymbol{X'X})^{-1}\boldsymbol{X'y}$.

Projection Matrix

- There is a geometric interpretation for the minimization problem that uses the idea of "projection".
- The following matrix is called a "projection matrix":

$$\boldsymbol{P} = \boldsymbol{X}(\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}'$$

- Intuition: $\boldsymbol{P}$ takes any vector in $\mathbb{R}^n$ and drops it onto the column space of $\boldsymbol{X}$—the closest point that can be written as $\boldsymbol{X}b$ for some b.
- The key payoff: $\boldsymbol{P}\boldsymbol{y} = \boldsymbol{X}\hat{\beta} = \hat{\boldsymbol{y}}$. This is why $\boldsymbol{P}$ is also called the "hat matrix"—it puts the hat on $\boldsymbol{y}$.

Properties of the Projection Matrix

- $\boldsymbol{P}$ leaves anything already in the column space alone:
  - $\boldsymbol{PX} = \boldsymbol{X}(\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}'\boldsymbol{X} = \boldsymbol{X}$
  - If $\boldsymbol{X} = [\boldsymbol{X}_1 \ \boldsymbol{X}_2]$, then $\boldsymbol{PX}_1 = \boldsymbol{X}_1$.
- $\boldsymbol{P}$ is **symmetric**: $\boldsymbol{P} = \boldsymbol{P}'$.
- $\boldsymbol{P}$ is **idempotent**: $\boldsymbol{PP} = \boldsymbol{P}$.
- These two properties (symmetric + idempotent) will do a lot of work for us in the rest of the course.

Projection onto column of 1

- If $\boldsymbol{X} = \mathbf{1}$, then $\boldsymbol{P}_1 = \mathbf{1}(\mathbf{1}'\mathbf{1})^{-1}\mathbf{1}' = \frac{1}{n}\mathbf{1}\mathbf{1}'$
- So if we project $\boldsymbol{y}$ onto $\boldsymbol{P}_1$, we get a vector repeating the sample mean:

$$\boldsymbol{P}_1\boldsymbol{y} = \frac{1}{n}\mathbf{1}\mathbf{1}'\boldsymbol{y} = \mathbf{1} \cdot \frac{1}{n}\sum_{i=1}^{n} Y_i = \mathbf{1}\bar{Y}$$

- **Takeaway:** The simplest regression (intercept only) projects $\boldsymbol{y}$ onto the sample mean. Every additional regressor refines this projection.
- The annihilator $\boldsymbol{M}_1\boldsymbol{y} = \boldsymbol{y} - \mathbf{1}\bar{Y}$ *demeans* $\boldsymbol{y}$—it removes the part explained by the intercept.

Orthogonal Projection

- The following matrix is called an "orthogonal projection matrix", or the Annihilator Matrix.

$$\begin{aligned} \boldsymbol{M} &= \boldsymbol{I}_n - \boldsymbol{P} \\ &= \boldsymbol{I}_n - \boldsymbol{X}(\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}' \end{aligned}$$

- $\boldsymbol{M}$ and $\boldsymbol{X}$ are orthogonal:

$$\begin{aligned} \boldsymbol{MX} &= (\boldsymbol{I}_n - \boldsymbol{P})\boldsymbol{X} \\ &= \boldsymbol{X} - \boldsymbol{PX} \\ &= \boldsymbol{X} - \boldsymbol{X} \\ &= 0 \end{aligned}$$

- We can define the residuals from projecting onto a column of 1s: $\boldsymbol{M}_1\boldsymbol{y} = \boldsymbol{y} - \mathbf{1}\bar{Y}$, this demeans $\boldsymbol{y}$.

Relationship between residuals and disturbances

$$\begin{aligned}
\hat{\boldsymbol{e}} &= \boldsymbol{y} - \hat{\boldsymbol{y}} \\
&= \boldsymbol{y} - \boldsymbol{X}\hat{\beta} \\
&= \boldsymbol{y} - \boldsymbol{P}\boldsymbol{y} \\
&= \boldsymbol{M}\boldsymbol{y} \\
&= \boldsymbol{M}(\boldsymbol{X}\beta + \boldsymbol{e}) \\
&= \boldsymbol{M}\boldsymbol{X}\beta + \boldsymbol{M}\boldsymbol{e} \\
&= \boldsymbol{M}\boldsymbol{e}
\end{aligned}$$

Projection and Residuals in R

```
mod <- lm(prestige ~ education + income, data=Prestige)
X <- cbind(1, Prestige$education, Prestige$income)
P <- X %*% solve(t(X) %*% X) %*% t(X)
M <- diag(nrow(X)) - P

# P*y = fitted values
all.equal(as.vector(P %*% y), fitted(mod))  # TRUE

# M*y = residuals
all.equal(as.vector(M %*% y), resid(mod))   # TRUE
```

$\boldsymbol{P}$ and $\boldsymbol{M}$ are used by R to compute `fitted()` and `resid()`.

Where We Stand

- We have $\hat{\beta} = (\boldsymbol{X'X})^{-1}\boldsymbol{X'y}$ and the geometry behind it ($\boldsymbol{P}$, $\boldsymbol{M}$).
- But how precise is $\hat{\beta}$? To answer that, we need $\text{Var}(\hat{\beta}|\boldsymbol{X}) = \sigma^2(\boldsymbol{X'X})^{-1}$.
- Problem: σ^2 is *unknown*—it's a population quantity, just like β.
- So we need a second estimator, s_e^2, built from the residuals. This is step 3.

Why Does Projection Help?

- We want to estimate the systematic part $\boldsymbol{X}\beta$. Two candidates:
  1. Use $\boldsymbol{y}$ itself (just report the raw data).
  2. Use $\boldsymbol{P}\boldsymbol{y} = \hat{\boldsymbol{y}}$ (the projection onto the column space of $\boldsymbol{X}$).
- Compare their variances as estimators of $\boldsymbol{X}\beta$:

$$\text{Var}(\boldsymbol{y}|\boldsymbol{X}) = \sigma^2 \boldsymbol{I}$$
$$\text{Var}(\boldsymbol{P}\boldsymbol{y}|\boldsymbol{X}) = \sigma^2 \boldsymbol{P}$$

- The difference is $\sigma^2(\boldsymbol{I} - \boldsymbol{P}) = \sigma^2 \boldsymbol{M}$, which is positive semi-definite.
- So $\text{Var}(\boldsymbol{y}) \geq \text{Var}(\boldsymbol{P}\boldsymbol{y})$: projection *removes noise* without distorting the signal, because $\boldsymbol{P}\boldsymbol{X}\beta = \boldsymbol{X}\beta$.

Variance of the regression errors

- We want to measure the precision of our regression estimates.
- The error (or disturbance) of an observation is the deviation of a value from its theoretical mean, cannot be observed.
- The coefficients inherit that uncertainty from the theoretical model.
- Our estimates have additional uncertainty, related to the fact we only have a sample.
- Residuals are the difference between observations and estimates, can be observed.

The Bane of Statistics: We Never See the Errors

	Disturbance e_i	**Residual** $\hat{e}_i$
Definition	$Y_i - \boldsymbol{x}_i'\beta$	$Y_i - \boldsymbol{x}_i'\hat{\beta}$
Depends on	population β	estimated $\hat{\beta}$
Observable?	No	Yes
Distribution	$E[e_i\|\boldsymbol{X}] = 0$, $\text{Var}(e_i) = \sigma^2$	$\hat{e}_i = \boldsymbol{M}_i\boldsymbol{e}$ (a linear combo of *all* e_j's)

- We showed that $\hat{\boldsymbol{e}} = \boldsymbol{M}\boldsymbol{e}$: each residual is a *filtered* version of the true disturbances, not a direct copy.
- This means MSE (computed on residuals) $\neq$ true error variance. We will need a degrees-of-freedom correction $(n-k)$ to fix this.

Estimating the Disturbance Variance: First Attempt

- Terminology note: *disturbance*, *error*, and *noise* all refer to $e_i = Y_i - \boldsymbol{x}_i'\beta$. We use them interchangeably; the key point is that e_i is **unobservable**.
- Natural idea: plug the residuals $\hat{e}_i$ into the sample variance formula:

$$\hat{\sigma}^2 = n^{-1}\hat{\boldsymbol{e}}'\hat{\boldsymbol{e}} = n^{-1}(\boldsymbol{My})'(\boldsymbol{My}) = n^{-1}\boldsymbol{y}'\boldsymbol{My} = n^{-1}\boldsymbol{e}'\boldsymbol{Me}$$

  where the last step uses $\boldsymbol{M} = \boldsymbol{M}'\boldsymbol{M}$ and $\boldsymbol{MX} = 0$.
- But $\hat{\boldsymbol{e}}'\hat{\boldsymbol{e}} \leq \boldsymbol{e}'\boldsymbol{e}$, so residuals underestimate the true sum of squared errors:

$$n^{-1}\boldsymbol{e}'\boldsymbol{e} - n^{-1}\boldsymbol{e}'\boldsymbol{Me} = n^{-1}\boldsymbol{e}'\boldsymbol{Pe} \geq 0$$

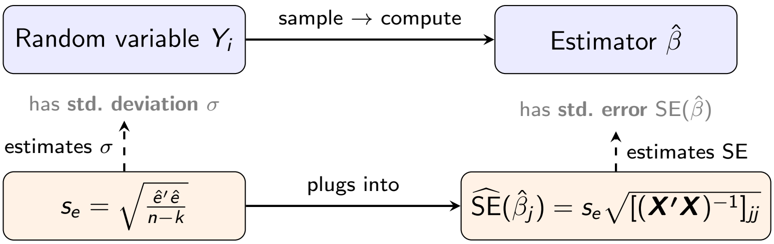
  because $\boldsymbol{P}$ is positive semi-definite.
- Projection "absorbs" some of the disturbance into fitted values, so residuals understate the true noise.

The Degrees-of-Freedom Correction

- Fix: divide by $n-k$ instead of n. Define $s_e^2 = \hat{\boldsymbol{e}}'\hat{\boldsymbol{e}}/(n-k)$.
- Proof that s_e^2 is unbiased for σ^2:

$$\begin{aligned}
\hat{\boldsymbol{e}}'\hat{\boldsymbol{e}} &= \boldsymbol{e}'\boldsymbol{M}'\boldsymbol{M}\boldsymbol{e} = \boldsymbol{e}'\boldsymbol{M}\boldsymbol{e} && (\boldsymbol{M}\text{ symmetric, idempotent})\\
E[\hat{\boldsymbol{e}}'\hat{\boldsymbol{e}}|\boldsymbol{X}] &= E[\boldsymbol{e}'\boldsymbol{M}\boldsymbol{e}|\boldsymbol{X}] && \\
&= E[\mathrm{tr}(\boldsymbol{e}'\boldsymbol{M}\boldsymbol{e})|\boldsymbol{X}] && (\text{scalar} = \text{its own trace})\\
&= E[\mathrm{tr}(\boldsymbol{M}\boldsymbol{e}\boldsymbol{e}')|\boldsymbol{X}] && (\mathrm{tr}(\boldsymbol{ABC})\text{=}\mathrm{tr}(\boldsymbol{CAB}))\\
&= \mathrm{tr}(\boldsymbol{M}\,E[\boldsymbol{e}\boldsymbol{e}'|\boldsymbol{X}]) && (\boldsymbol{M}\text{ fixed given }\boldsymbol{X})\\
&= \mathrm{tr}(\boldsymbol{M}\sigma^2\boldsymbol{I}) = \sigma^2\mathrm{tr}(\boldsymbol{M}) = \sigma^2(n-k) && (\text{A4, }\mathrm{tr}(\boldsymbol{M})\text{=}n-k)
\end{aligned}$$

Standard Deviation vs. Standard Error



- **Standard deviation**: spread of a random variable (σ).
- **Standard error**: spread of an *estimator* across repeated samples.
- The punchline: to estimate how uncertain $\hat{\beta}$ is, we first need to estimate how noisy Y is. Estimation all the way down.

Interpretation of Residual Standard Error

- summary(lm()) reports s_e as the Residual standard error or "sigma"
- STATA reports this quantity as "Root Mean Squared Error".
- s_e is on the same scale as the outcome.
- If $y > 0$, it can make sense to calculate an *average prediction error rate*: $\frac{s_e}{\bar{y}}$.
- Example: if we are predicting houses that are on average 1 million dollars, being generally off by $50k is less bad than if we are predicting houses that are $100k.

What Are Standard Errors For?

Consider a regression of occupational prestige on education and income:

	Estimate	Std. Error	t value	$\Pr(>\lvert t\rvert)$
(Intercept)	-6.06	4.27	-1.42	0.159
education	4.19	0.39	10.78	< 0.001
income	0.0013	0.0003	5.19	< 0.001

Residual std. error: $s_e = 7.81$ on 99 d.f. $R^2 = 0.80$

- The Std. Error column tells us how much $\hat{\beta}_j$ would vary across repeated samples.
- Education: estimate $= 4.19$, SE $= 0.39$. The estimate is about 10 standard errors from zero—hard to explain by sampling noise alone.
- Where does each SE come from? From s_e and $(\boldsymbol{X}'\boldsymbol{X})^{-1}$. Let's derive that.

Variance of $\hat{\beta}$

$$
\begin{aligned}
V(\hat{\beta}|\boldsymbol{X}) &= V((\boldsymbol{X'X})^{-1}\boldsymbol{X'y}|\boldsymbol{X}) && \text{(Normal Equation)} \\
&= V(\boldsymbol{Ay}|\boldsymbol{X}) && \text{(Pick } \boldsymbol{A} \text{ as placeholder)} \\
&= \boldsymbol{A}V(\boldsymbol{y}|\boldsymbol{X})\boldsymbol{A}' && (\text{Var}(cX) = c^2\text{Var}(X)) \\
&= \boldsymbol{A}\Sigma\boldsymbol{A}' \\
&= \boldsymbol{A}(\sigma^2\boldsymbol{I})\boldsymbol{A}' && \text{(Assumption A4)} \\
&= \sigma^2\boldsymbol{AA}' \\
&= \sigma^2(\boldsymbol{X'X})^{-1}\boldsymbol{X'X}(\boldsymbol{X'X})^{-1} \\
&= \sigma^2(\boldsymbol{X'X})^{-1} \\
\hat{V}(\hat{\beta}|\boldsymbol{X}) &= s_e^2(\boldsymbol{X'X})^{-1} && \text{(Standard Error of } \hat{\beta}\text{)}
\end{aligned}
$$

Standard Errors in R: Connecting to the Formula

```r
mod <- lm(prestige ~ education + income, data=Prestige)

# The formula from the slides: s^2 (X'X)^{-1}
sigma(mod)^2 * solve(t(X) %*% X)

# What R reports:
vcov(mod)

# Standard errors = sqrt of the diagonal
sqrt(diag(vcov(mod)))
coef(summary(mod))[, "Std. Error"]  # identical
```

Every standard error in `summary(lm())` comes from $\sqrt{\text{diag}(s_e^2(\boldsymbol{X'X})^{-1})}$.

Splitting $\boldsymbol{y}$ into Signal and Noise

- Every observation is part signal, part noise. Projection lets us split them:

$$\boldsymbol{y} = \underbrace{\boldsymbol{P}\boldsymbol{y}}_{\text{signal } \hat{\boldsymbol{y}}} + \underbrace{\boldsymbol{M}\boldsymbol{y}}_{\text{noise } \hat{\boldsymbol{e}}}$$

- The two pieces are **orthogonal**: $\hat{\boldsymbol{y}}'\hat{\boldsymbol{e}} = (\boldsymbol{P}\boldsymbol{y})'(\boldsymbol{M}\boldsymbol{y}) = \boldsymbol{y}'\boldsymbol{P}\boldsymbol{M}\boldsymbol{y} = 0$.
- So their squared lengths add up (Pythagorean theorem in $\mathbb{R}^n$):

$$\|\boldsymbol{y}\|^2 = \|\hat{\boldsymbol{y}}\|^2 + \|\hat{\boldsymbol{e}}\|^2$$
$$\sum_{i=1}^{n} Y_i^2 = \sum_{i=1}^{n} \hat{Y}_i^2 + \sum_{i=1}^{n} \hat{e}_i^2$$

- How much of the total length is signal? That question leads to R^2.

Analysis of Variance

- Subtract $\bar{Y}$ from both sides of the decomposition:

$$\boldsymbol{y} - \mathbf{1}\bar{Y} = (\hat{\boldsymbol{y}} - \mathbf{1}\bar{Y}) + \hat{\boldsymbol{e}}$$

- Take inner products. The cross terms vanish when $\boldsymbol{X}$ contains a constant:

$$(\hat{\boldsymbol{y}} - \mathbf{1}\bar{Y})'\hat{\boldsymbol{e}} = \hat{\boldsymbol{y}}'\hat{\boldsymbol{e}} - \bar{Y}\mathbf{1}'\hat{\boldsymbol{e}} = 0 - 0 = 0$$

- So we get the **analysis of variance** (ANOVA) decomposition:

$$\|\boldsymbol{y} - \mathbf{1}\bar{Y}\|^2 = \|\hat{\boldsymbol{y}} - \mathbf{1}\bar{Y}\|^2 + \|\hat{\boldsymbol{e}}\|^2$$

$$\underbrace{\sum (Y_i - \bar{Y})^2}_{\text{SST}} = \underbrace{\sum (\hat{Y}_i - \bar{Y})^2}_{\text{SSR (regression)}} + \underbrace{\sum \hat{e}_i^2}_{\text{SSE (error)}}$$

Goodness-of-Fit: R^2

- Dividing through by SST:

$$1 = \frac{\text{SSR}}{\text{SST}} + \frac{\text{SSE}}{\text{SST}}$$
$$R^2 \equiv \frac{\text{SSR}}{\text{SST}} = 1 - \frac{\text{SSE}}{\text{SST}}$$

- R^2 is the proportion of the variance in Y that is linearly explained by $\boldsymbol{X}$.
- $0 \leq R^2 \leq 1$: equals 0 when $\boldsymbol{X}$ explains nothing; equals 1 when $\hat{e}_i = 0$ for all i.
- We can always make $R^2 = 1$ by adding enough linearly independent columns to $\boldsymbol{X}$ (one per observation). This does not mean the model is good.

R^2 in Practice

- Typical values of R^2 vary by field:
  - Cross-sectional micro data (e.g. earnings regressions): $R^2 \approx 0.2$–0.4
  - Aggregate time-series (e.g. GDP growth models): $R^2 \approx 0.7$–0.9
  - Experimental data with noisy outcomes: R^2 can be very low and the treatment effect still precisely estimated.
- In R: `summary(lm(y ~ x))$r.squared` reports R^2.
- The **adjusted** $R^2 = 1 - \frac{n-1}{n-k}(1 - R^2)$ penalizes adding regressors and can decrease when a variable adds no explanatory power.

Naming Conventions: A Warning

- Different textbooks use SSE and SSR with *opposite* meanings:

	Hansen / this course	**Some other texts**
$\sum(\hat{Y}_i - \bar{Y})^2$	SSR (regression)	ESS or SSR
$\sum \hat{e}_i^2$	SSE (error)	RSS or SSE

- The underlying math is always the same: $\text{SST} = (\text{explained}) + (\text{unexplained})$.
- When reading papers or other textbooks, check which convention is in use.

The Cauchy–Schwarz Inequality

Cauchy–Schwarz Inequality

For any two random variables U and V with finite variances:

$$\left|\text{Cov}(U, V)\right| \;\leq\; \text{sd}(U)\,\text{sd}(V)$$

with equality if and only if $V = aU + b$ for some constants a, b.

- Dividing both sides by $\text{sd}(U)\,\text{sd}(V)$ gives

$$-1 \;\leq\; \text{corr}(U, V) \;\leq\; 1$$

  with $|\text{corr}(U, V)| = 1$ only if the relationship is perfectly linear.
- This is an **algebraic fact**, not a statistical assumption. It holds for *any* joint distribution.
- Implication for the BLP slope: since $\beta = \text{Cov}(X, Y)/\text{Var}(X)$, we can write

Cauchy–Schwarz Applied: When $\mathrm{Var}(Y) = \mathrm{Var}(X)$

- Suppose Y and X have the **same marginal variance**: $\mathrm{Var}(Y) = \mathrm{Var}(X)$.
- Then the BLP slope simplifies:

$$\beta = \frac{\mathrm{Cov}(X, Y)}{\mathrm{Var}(X)} = \frac{\mathrm{Cov}(X, Y)}{\mathrm{sd}(X)\,\mathrm{sd}(Y)} = \mathrm{corr}(X, Y)$$

- By Cauchy–Schwarz: $|\beta| \leq 1$, with equality only if $Y = aX + b$.
- If we also assume $\mathbb{E}[Y] = \mathbb{E}[X] = \mu$ (same mean), then

$$\alpha = \mu - \beta\mu = (1 - \beta)\mu$$

so the BLP becomes:

Regression to the Mean

$$P(Y \mid X) = \mu + \beta(X - \mu), \qquad |\beta| < 1$$

Regression to the Mean: The Setup

- Francis Galton (1886) measured fathers' and sons' heights. Tall fathers had sons who were tall—but *less* tall than their fathers. Short fathers had sons who were short—but *less* short.
- This is a natural result of the equal-variance projection we just derived: when $\text{Var}(Y) = \text{Var}(X)$, the slope is the correlation, and $|\beta| < 1$.
- The same structure appears in sample quantities. We proved $\text{SST} = \text{SSR} + \text{SSE}$. Divide by n:

$$\frac{1}{n}\sum(Y_i - \bar{Y})^2 = \frac{1}{n}\sum(\hat{Y}_i - \bar{Y})^2 + \frac{1}{n}\sum \hat{e}_i^2$$

$$\text{Var}(Y) = \text{Var}(\hat{Y}) + \text{Var}(\hat{e})$$

- So the variance of the fitted values is *smaller* than the variance of the raw data—some variance is "left behind" in the residuals.

Regression to the Mean: How Much Shrinkage?

- How much smaller? Recall $R^2 = \text{SSR}/\text{SST}$, so $\text{SSR} = R^2 \cdot \text{SST}$:

$$\frac{1}{n}\sum(\hat{Y}_i - \bar{Y})^2 = R^2 \cdot \frac{1}{n}\sum(Y_i - \bar{Y})^2 \qquad \text{(divide both sides by } n\text{)}$$

$$\text{Var}(\hat{Y}) = R^2 \cdot \text{Var}(Y)$$

- When $R^2 < 1$, fitted values have **less spread** than the raw data—they are shrunk toward $\bar{Y}$. The noisier the relationship, the more shrinkage.
- Example: if $R^2 = 0.5$, the fitted values have half the variance of the data.

The Regression Fallacy

- We showed $\beta < 1$ under the assumption that $\text{Var}(Y) = \text{Var}(X)$ (a *stable* distribution).
- **Common error:** observing $\beta < 1$ and concluding that $\text{Var}(Y) < \text{Var}(X)$—that the distribution is *converging*.
- But from the projection $Y = \alpha + \beta X + e$ with $\text{Cov}(X, e) = 0$:

$$\text{Var}(Y) = \beta^2 \text{Var}(X) + \text{Var}(e)$$

  $\text{Var}(Y) < \text{Var}(X)$ requires $\beta^2 < 1 - \text{Var}(e)/\text{Var}(X)$, which is much stronger than $|\beta| < 1$.
- The variance of the projection error e **replenishes the spread** that regression to the mean appears to remove.

Examples of Regression to the Mean

- A student scores in the 95th percentile on Midterm 1, then the 80th on Midterm 2. Did they get lazy?
    - The "Sports Illustrated jinx" (athletes on the cover then underperform)
    - "Sophomore slumps" in politics, sports, music
    - Why the worst-performing schools improve and the best-performing schools decline, even without any intervention
- In 1933, Horace Secrist published *The Triumph of Mediocrity in Business*, documenting that department stores sorted by 1920 profits "converged toward mediocrity" over the next decade. There was no discovery—regression to the mean is a necessary feature of stable distributions.

Why This Matters for Research

- Regression to the mean is a threat to any research design that selects on the outcome or a noisy proxy of it:
  - Evaluating a tutoring program given to students who failed a test
  - Assessing a policy intervention targeted at the worst-performing hospitals
  - Studying whether "crackdowns" reduce crime in high-crime areas
- In each case, the group would have improved on average *even without the treatment*, because you selected them at an extreme.
- This is why we need control groups, and why "before–after" comparisons on selected samples are unreliable.